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## Design And Analysis Of Algorithms [BCSE204L] CAT-1 A2 2024-2025

| Q.No | Question                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | M | CO  | BL  |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|-----|-----|
| 1.   | <p>a. Design an algorithm by finding the required design paradigm to find the highest and lowest marks scored by students in the exam by fulfilling the below parameters and derive the respective recurrence relation for the same.</p> <p>i. Base case 1(If the input is only one element) give solution i.e what is highest and lowest?</p> <p>ii. Base case 2(If the input is two elements) give solution.</p> <p>iii. If the input is more than two elements, divide the inputs into sub-problems and use the recursive function for solving the sub-problems and write the procedure for combining the sub-solutions.</p> <p><b>Solution: Divide and Conquer Strategy</b></p> <pre>Algorithm MaxMin(i, j, max, min) // a[i..j] is an array. Parameters i and j are integers. // 1 ≤ i ≤ j ≤ n. The effect is to set max and min to the // largest and smallest values in a[i..j], respectively. {     If (i == j) then max := min := a[i]; // Smallest case of Small(P)     else if (i == j - 1) then // Another case of Small(P)     {         If (a[i] &lt; a[j]) then         {             max := a[j]; min := a[i];         }         else         {             max := a[i]; min := a[j];         }     }     else     {         // If P is not small, divide P into subproblems.         // Find where to split the set.         mid := (i + j) / 2;         // Solve the subproblems.         MaxMin(i, mid, max1, min1);         MaxMin(mid + 1, j, max2, min2);         // Combine the solutions.         If (max &lt; max1) then max := max1;         If (min &gt; min1) then min := min1;     } }</pre> | 5 | C01 | BL4 |

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ANSWER KEY



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## Design And Analysis Of Algorithms [BCSE204L] CAT-1 A2 2024-2025

|    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |    |     |     |
|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-----|-----|
| 2. | <p>Demonstrate how the Huffman code is used to make data smaller by Compressing it? Consider the following string "BCAADDCCACAC" compress it and find out how many minimum numbers of bits are required for transmitting the above said string. Write down the needed process and the respective algorithm and analyse the time complexity.(5M)</p> <p><b>Algorithm:</b></p> <pre>Huffman(C) 1 x = C 2 Q = C 3 for i = 1 to x - 1 4     allocate a new node z 5     z.left = x = EXTRACT-MIN(Q) 6     z.right = y = EXTRACT-MIN(Q) 7     z.freq = x.freq + y.freq 8     INSERT(Q, z) 9 return EXTRACT-MIN(Q) // return the root of the tree</pre> <p><b>Time Complexity : <math>O(n \log n)</math></b></p> <p>i. Calculate the frequency of each character(1M)<br/><b>A: 5 times B: 1 time C: 6 times D: 3 times</b></p> <p>ii. Build a priority Queue(or min heap) (1M)<br/>Priority Queue :<br/><b>B: 1 D: 3 A: 5 C: 6</b></p> <p>iii. Build the Huffman Tree(1M)</p> | 10 | C01 | BL3 |
|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-----|-----|

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|    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |    |     |     |
|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-----|-----|
| 3. | <p>Given an array consisting of number of following elements {-2, -5, 6, -2, -3, 1, 5, -6}. Find a contiguous non-empty subarray within the array that has the maximum sum, and return the sum as well as the subarray elements. Implement using appropriate design paradigm and give the optimized algorithm and derive the time complexity.</p> <p><b>Max Subarray Sum: 7</b></p> <p><b>Subarray Elements are: {6,-2,-3,1,5}</b></p> <p><b>Solution: Divide and Conquer Strategy</b></p> <pre>FIND-MAXIMUM-SUBARRAY(A, low, high) 1  if high == low 2  return (low, high, A[low]) // base case: only one element 3  else mid = (low + high) / 2 4  (left-low, left-high, left-sum) =     FIND-MAXIMUM-SUBARRAY(A, low, mid) 5  (right-low, right-high, right-sum) =     FIND-MAXIMUM-SUBARRAY(A, mid + 1, high) 6  (cross-low, cross-high, cross-sum) =     FIND-MAX-CROSSING-SUBARRAY(A, low, mid, high) 7  if left-sum &gt; right-sum and left-sum &gt; cross-sum 8  return (left-low, left-high, left-sum) 9  elseif right-sum &gt; left-sum and right-sum &gt; cross-sum 10 return (right-low, right-high, right-sum) 11 else return (cross-low, cross-high, cross-sum)</pre> | 10 | C01 | BL4 |
|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-----|-----|

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Design And Analysis Of Algorithms [BCSE204L] CAT-1 A2 2024-2025

|    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |    |     |     |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-----|-----|
| 4. | <p>Consider a knapsack bag where <math>n = 4</math>, and the values (profits) are given by {10, 12, 12, 18} and the weights given by {2, 4, 6, 9}. The maximum weight is given by <math>W = 15</math>. The solution is to find subset of objects such that the total value is maximized, and the sum of weights of the objects does not exceed a given capacity <math>W</math>. Implement and write the algorithm using Dynamic Programming strategy and derive time complexity.</p> <p><b>Solution: Dynamic Programming</b></p> <p><b>Max Profit: 40</b></p> <p><b>Weights Considered: {1,1,0,1}</b></p> <p><b>Formula for Filling the Table: <math>V[i,j] = \text{Max}\{v[i-1,j], v[i-1,j-w_i] + v_i\}</math></b></p> <p><b>Time Complexity:</b></p> <ul style="list-style-type: none"><li>Time for constructing the table: <math>O(nW)</math></li><li>Time taken for Composition of Optimal load: <math>O(n+W)</math></li></ul> | 10 | C02 | BL1 |
| 5. | <p>i. In real time application, many problems which deals with searching for set of solutions or which ask for an optimal solution satisfying some constraints can be solved using which design paradigm? Define the Design paradigm and Differentiate implicit and explicit Constraints with example.</p> <p><b>Solution: Backtracking</b></p> <p><b>Definition:</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 10 | C02 | BL2 |

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## Design And Analysis Of Algorithms [BCSE204L] CAT-1 A1 2023-2024

**Q.No.**      **Questions**

- a) Discuss the selection sort algorithm by providing its pseudo code; Discuss the loop invariant in connection with selection sort algorithms Check the proof of correctness for the same. (6)

Pseudo code of Selection sort : (3)

- i) Loop Invariant, Initialization, Maintenance, and Termination (2M)

b) Using the master's theorem, solve the recurrence relation  
 i)  $T(n) = n^2 \left[ \frac{7}{8} \right] + c'$   
 ii)  $T(n) = 7T\left(\frac{n}{8}\right) - 16n^2$  (4M)  
 (3 M) each  
 iii)  $T(n) = O(n^2 \log n) + n^2 \Rightarrow T(n) = O(n^2 \log n) + O(n^2)$   
 iv)  $T(n) = 7T\left(\frac{n}{8}\right) + 10n \Rightarrow T(n) = O(n^{2.809}) + 10n$

- c) Consider the string "abacabadcdcdcbabcd". Use minimum number of bits for transmitting the said string. Calculate the number of bits used to encode this using Huffman coding technique. Identify the bits required by both fixed-size and variable length encoding. (10)

Huffman tree : (2 M)  
Huffman code table : (3 M)  
Bits for fixed size coding : (1 M)  
Bits for variable coding : (3 M)

| Character | Freq. |
|-----------|-------|
| a         | 1     |
| b         | 1     |
| c         | 2     |
| d         | 2     |
| e         | 4     |

Q 11  
Q 12  
Q 13  
Q 14

0 1  
1 1  
1 1  
1 1

10  
6 3  
3 10  
1 10  
0 2  
10 10  
10 10

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## Design And Analysis Of Algorithms [BCSE204L] CAT-1 A1 2023-2024

ANS:

M matrix (5 marks)  
S matrix (2 marks)  
Optimal parenthesization (2 marks)

M(1|2) = 100  
M(2|3) = 250  
M(2|4) = 300  
M(3|5) = 300  
M(1|2) = 200  
M(2|3) = 400  
M(1|3) = 400  
M(1|4) = 200  
M(2|5) = 700  
M(1|5) = 300  
The solution is 100  
Optimal parenthesization is ((AB)C)DE

5. Define 4-Queens problem. Assuming that the queens are placed column-wise in the 4-Queens problem, solved using backtracking. Consider the following intermediate array, where the queens are attacking each other. Show the steps that involve backtracking to attain the solution where no queens attack the other.

|   |   |   |   |
|---|---|---|---|
| Q |   |   |   |
|   | Q |   |   |
|   |   | Q |   |
|   |   |   | Q |

Each step where backtracking is done must be shown along with the necessary diagrams.

Ans:

Definition: (4 m)  
Backtracking: (8 m)  
Students MUST follow COLUMNS wise working out, not ROW wise.  
Backtrack to 3<sup>rd</sup> column, then follow working out.  
The final answer must be

|   |   |   |   |
|---|---|---|---|
|   |   | Q |   |
| Q |   |   |   |
|   | Q |   |   |
|   |   |   | Q |

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## Design And Analysis Of Algorithms [BCSE204L] CAT-1 A2 2023-2024

VIT  
Vellore Institute of Technology

School of Computer Science and Engineering

Winter Semester 2023-2024

Continuous Assessment Test - 1

Programme Name & Branch: B.Tech (BC/BCCE/BC/ICT/IDS/SEKT) Set : A2-TA2

Course Name & code: BCSE204L – Design and Analysis of Algorithms

Class Number (s): ALL

Faculty Name (s): ALL

Exam Duration: 90 Mins. Max. Marks: 80

General instructions: ANSWER ALL THE QUESTIONS

| Q.No. | Question                                                                                                                                                                                                                                                                                                                                                                                                                                                | Max. Marks | CO  | BL  |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----|-----|
| 1.    | a) Demonstrate the iteration method to compute the asymptotic complexity for the following recurrence:<br>$T(n) = 4T\left(\frac{n}{2}\right) + n^2$<br>b) Use master's method to compute the asymptotic complexity for the following recurrence. In each case, identify the case of master method that it uses to compute the asymptotic complexity:<br>(a) $T(n) = 16T\left(\frac{n}{4}\right) + n^2$<br>(b) $T(n) = 9T\left(\frac{n}{3}\right) + n^2$ | 10         | CO1 | BL5 |
| 2.    | Discuss how the greedy approach is used to solve optimization problems. Contrast the frequency table of characters in "HELLO there are you?" in a non-decreasing order of frequency. Use Huffman code to find the code word for each character.                                                                                                                                                                                                         | 10         | CO2 | BL3 |
| 3.    | Define maximum sub-array sum problem. Find the series of contiguous elements that results in the maximum sub-array sum for the array given below:<br>[-2, -3, 1, 4, -2, 3, 15, -4, 6, -1]                                                                                                                                                                                                                                                               | 10         | CO2 | BL4 |
| 4.    | Longest common subsequence (LCS) problem is the problem of finding the longest subsequence common to all sequences in a set of sequences. Consider the sequences "ACCGGTCGAGT" and "ATCCTGTCG". Find the length of the longest common subsequence using dynamic programming approach with the pseudocode for the same.                                                                                                                                  | 10         | CO2 | BL5 |
| 5.    | Given a set of non-negative integers S = {1, 34, 4, 12, 5, 2} and a sum 36, determine the subsets of S, whose sum is equal to 36 using backtracking.                                                                                                                                                                                                                                                                                                    | 10         | CO2 | BL4 |

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## Design And Analysis Of Algorithms [BCSE204L] CAT-1 A1 2024-2025

**VIT**  
Vellore Institute of Technology

REG.NO.:

NAME OF THE SCHOOL:  
CONTINUOUS ASSESSMENT TEST - 1  
WINTER SEMESTER 2024-2025  
SLOT: A1+A3

Programme Name & Branch : B.Tech CSE  
Course Code and Course Name : BCSE204L / Design and Analysis of Algorithms  
Faculty Name(s) : ALL  
Class Number(s) : ALL  
Date of Examination : 27.1.25  
Exam Duration : 90 minutes  
Maximum Marks: 50

General Instruction(s):

- Answer All Questions

| Q.No | Question                                                                                                                                                                                                                                                                                               | M  | CO | BL |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|
| 1.   | "Securityingtolearndesignandanalysisofalgorithms" for the given phrase construct the Huffman coding and generate the code for "Tester". Compare the length of code between fixed and variable length code for the given word "Tester".                                                                 | 10 | 1  | 2  |
| 2.   | Develop an algorithm using divide and conquer strategy to multiply two integers 38499 and 83329. Generate the recurrence relation from the algorithm and compute the time complexity. Also compute how much time it will take to multiply the given two integers.                                      | 10 | 1  | 3  |
| 3.   | a) Apply master theorem to calculate the time complexity of the recurrence relation $T(n) = 2T(n/4) + \sqrt{n} + 42$ . Illustrate the evaluation step by steps.<br>b) Apply backward substitution to enable the time complexity for the recurrence relation $T(n) = 2T(n/2) + n \log n$ .              | 4  | 1  | 3  |
| 4.   | Given a chain of matrix:<br>A1: 5X4<br>A2: 4X3<br>A3: 3X5<br>A4: 5X2<br>A5: 2X4<br>where A <sub>i</sub> has dimensions p <sub>i</sub> x q <sub>i</sub> , fully parenthesize the product A <sub>1</sub> A <sub>2</sub> ... A <sub>n</sub> in a way that minimizes the number of scalar multiplications. | 10 | 2  | 1  |
| 5.   | Given a set of positive integers {3,2,5,7,6,8,10,4} and a target value "20", determine whether some subset has a sum equal to the target.                                                                                                                                                              | 10 | 2  | 3  |

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## Design And Analysis Of Algorithms [BCSE204L] CAT-1 G2 2022-2023

**VIT**  
Vellore Institute of Technology

School of Computer Science and Engineering

Winter Semester 2022-2023

Continuous Assessment Test - 1

Programme Name & Branch: B.Tech (BC/BCE/BCU/BCU/BCU/BCU) Slot : G2-TG2  
Course Name & code: BCSE204L - Design and Analysis of Algorithms  
Class Number (s): ALL  
Faculty Name (s): 18622/18781/11226/12357/12477/19780/14713/14816/14411/16780/17970/18807/18811/18871/18872/18876/18938/19084/19570/19586/19614/19620/19662/13329/18877  
Exam Duration: 90 Min. Max. Marks: 50

General Instruction(s): ANSWER ALL THE QUESTIONS

| Q.No | Question                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Max Marks | CO  | BL   |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-----|------|
| 1.   | a) Which one of the following is the recurrence equation for the worst-case time complexity of the MergeSort algorithm for sorting n (32) numbers in the recurrence equation given in the options below? c is a constant.<br>i. $T(n) = 2T(n/2) + c$<br>ii. $T(n) = T(n/1) + T(n) + cn$<br>iii. $T(n) = 2T(n/1) + cn$<br>iv. $T(n) = T(n/2) + n$<br>Justify your answer and solve the equation using appropriate method. (2+4 Marks)<br>b) Using the master's theorem, solve the recurrence relation $T(n) = 7T(n/2) + n^2$ . (6 Marks) | 10        | CO1 | BL.5 |
| 2.   | Design and develop an algorithm to multiply 2 integers and analyze their time complexity. Explain the technique used to speed up the computation using the numbers 1234 and 4321.                                                                                                                                                                                                                                                                                                                                                       | 10        | CO2 | BL.3 |
| 3.   | Write the optimal cost function using dynamic programming and give the detailed discussion for the following scenario. Given a set of non-negative integers, and a value sum, determine if there is a subset of the given set with sum equal to the given sum.                                                                                                                                                                                                                                                                          | 10        | CO2 | BL.4 |
| 4.   | Elaborate the optimal substructure property of Huffman coding using appropriate argument. Construct the frequency table of characters in "mississippi" in non-decreasing order of frequency. Using Huffman code method, find the code word for each character.                                                                                                                                                                                                                                                                          | 10        | CO2 | BL.5 |
| 5.   | The N-Queens (e.g. N=4), problem is to place the N chess queens on an N×N chessboard so that no two queens attack each other. Discuss with an example and solve it using backtracking.                                                                                                                                                                                                                                                                                                                                                  | 10        | CO2 | BL.4 |

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## Design And Analysis Of Algorithms [BCSE204L] CAT-1 G1 2022-2023

| Q.No | Question                                                                                                                                                                                                                                                                                                                                                                                                   | Max. Marks | CO  | BL  |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----|-----|
| 1.   | a) Write the iterative algorithm to determine the product of first 'n' natural numbers. Prove its correctness using loop invariant method. (5 Marks)<br>b) Consider the following pseudo code<br><pre>func(x, n) {     if (n==0)         return 1;     if (n==1)         return x;     else         return (x * func(x, n-1)); }</pre> Analyse its functionality and derive its time complexity. (5 Marks) | 10         | CO1 | BL4 |
| 2.   | a) Write the pseudo code for constructing the Huffman tree using greedy method. (4 Marks)<br>b) Consider the string "abccddeeff". Each letter in the string must be assigned a binary code. Generate the code word for each character by constructing the Huffman tree. (6 Marks)<br>c) Find the minimum length of encoded string.                                                                         | 10         | CO2 | BL3 |
| 3.   | Consider a chain of 5 matrices and the matrix dimensions as follows:<br>Matrix: $A_1$ $A_2$ $A_3$ $A_4$ $A_5$                                                                                                                                                                                                                                                                                              | 10         | CO3 | BL3 |

| Dimensions | 4 x 10                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 10 x 3 | 3 x 12 | 12 x 20 | 20 x 7 |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|--------|---------|--------|
| 1.         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |        |        |         |        |
| 4.         | Apply dynamic programming strategy to find the minimum number of scalar multiplication to compute the product of matrices $A_1, A_2, A_3, A_4, A_5$ with optimal parenthesization. The subarray sum $S(i, j)$ is the contiguous subarray with the largest sum. Consider a sequence of 8 elements: $A = \{-2, 3, 4, -1, -2, 1, 5, -1\}$ . Design (Design) algorithm to find the maximum sum subarray and determine the $S(i, j)$ of given $A$ , where $1 \leq i \leq j \leq 8$ . |        |        |         |        |
| 5.         | Graph coloring is the procedure of assignment of colors to each vertex of a graph such that no adjacent vertices get same color. Design an algorithm for this problem by applying a backtracking method and colour the below graph using red, yellow, and blue (m=3) (show any 10 possible solutions in state space tree).                                                                                                                                                      |        |        |         |        |

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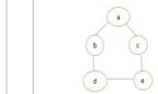


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| Dimensions | 4 x 10                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 10 x 3 | 3 x 12 | 12 x 20 | 20 x 7 |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|--------|---------|--------|
| 1.         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |        |        |         |        |
| 4.         | Apply dynamic programming strategy to find the minimum number of scalar multiplication to compute the product of matrices $A_1, A_2, A_3, A_4, A_5$ with optimal parenthesization. The subarray sum $S(i, j)$ is the contiguous subarray with the largest sum. Consider a sequence of 8 elements: $A = \{-2, 3, 4, -1, -2, 1, 5, -1\}$ . Design (Design) algorithm to find the maximum sum subarray and determine the $S(i, j)$ of given $A$ , where $1 \leq i \leq j \leq 8$ . |        |        |         |        |
| 5.         | Graph coloring is the procedure of assignment of colors to each vertex of a graph such that no adjacent vertices get same color. Design an algorithm for this problem by applying a backtracking method and colour the below graph using red, yellow, and blue (m=3) (show any 10 possible solutions in state space tree).                                                                                                                                                      |        |        |         |        |



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